

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

$f(x) = x + b$

For the linear function f , b is a constant. When $x = 0$, $f(x) = 30$. What is the value of b ?

Answer

- A. -30
- B. $-\frac{1}{30}$
- C. $\frac{1}{30}$
- D. 30

Correct Answer: D

Rationale

Choice D is correct. It's given that when $x = 0$, $f(x) = 30$. Substituting 0 for x and 30 for $f(x)$ in the given function yields $30 = 0 + b$, or $30 = b$. Therefore, the value of b is 30 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.